

$Q-P$

$\lambda \in \mathbb{R}$

$v \in \mathbb{R}^2, \mathbb{R}^3$

λv

$v=(x,y)$ | $v=(x,y,z)$

$\lambda v=(\lambda x, \lambda y)$ | $\lambda v=(\lambda x, \lambda y, \lambda z)$

Prodotto scalare

$\mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$

$\mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}$

$(x_1,y_1)\cdot(x_2,y_2)=x_1x_2+y_1y_2 \in \mathbb{R}$

$(\vec{v},\vec{w})$ $\quad$ $\langle , \rangle$ $\quad$ $(x_1,y_1)\begin{pmatrix}x_2\\y_2\end{pmatrix}$

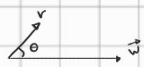
$v \cdot w$ $\quad$ $v^T w$

(v,w) $\quad$ $\langle v,w \rangle$

$(x_1,y_1,z_1)\cdot(x_2,y_2,z_2)=x_1x_2+y_1y_2+z_1z_2$

$v=(x_1,y_1,z_1)$

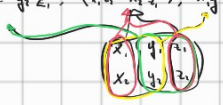
$w=(x_2,y_2,z_2)$

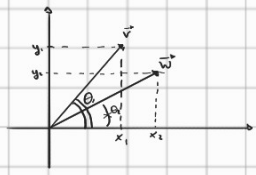


$v \cdot w = |v| \cdot |w| \cdot \cos\theta$

$v \times w$ o $v \wedge w \in \mathbb{R}^3$ (wedge)

$(y_1z_2-y_2z_1, -(x_1z_2-x_2z_1), x_1y_2-x_2y_2)$





$x_1 = |v| \cdot \cos\theta_1$

$y_1 = |v| \cdot \sin\theta_1$

$x_2 = |\vec{w}| \cdot \cos\theta_2$

$y_2 = |\vec{w}| \cdot \sin\theta_2$

$v \cdot w =$

$= x_1x_2 + y_1y_2 =$

$= |v| \cdot \cos\theta_1 \cdot |w| \cdot \cos\theta_2 + |v| \sin\theta_1 \cdot |w| \sin\theta_2 =$

$= |v| \cdot |w| \cdot (\cos\theta_1 \cdot \cos\theta_2 + \sin\theta_1 \cdot \sin\theta_2) =$

$= |v| \cdot |w| \cdot \cos(\theta_1 - \theta_2)$

$\forall v_1, w_1, w_2 \in \mathbb{R}^2, \mathbb{R}^3$

$\forall \lambda, \mu \in \mathbb{R}$

$v \cdot (\lambda w_1 + \mu w_2) =$

$= \lambda (v \cdot w_1) + \mu (v \cdot w_2)$

$\forall v, w \in \mathbb{R}^2, \mathbb{R}^3$

$v \cdot w = w \cdot v$

$v \perp w$ $\quad$ v e w sono perpendicolari $\Leftrightarrow$ $v \cdot w = 0$

$(v \wedge w) \cdot v = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix} = 0$ (determinante)

$v \wedge w \perp v$

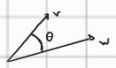
$v \wedge w \perp w$

$\forall v, w \in \mathbb{R}^3$ $\quad \forall \lambda, \mu \in \mathbb{R}$

$v \wedge (\lambda w_1 + \lambda w_2) = \lambda (v \wedge w_1) + \mu (v \wedge w_2)$

$\forall v, w \in \mathbb{R}^3$ $\quad v \wedge w = -w \wedge v$

$|v \wedge w| = |v| \cdot |w| \cdot \sin\theta \longrightarrow |v \wedge w|^2 = |v|^2 |w|^2 - (v \cdot w)^2$ (Identità di Lagrange)



$v \wedge w = 0 \Leftrightarrow v \parallel w$ vettori paralleli o proporzionali



$S = \frac{1}{2} |(B-A) \wedge (C-A)|$



"Prodotto misto" di v_1, v_2, v_3

$v_1 (v_2 \wedge v_3)$ $\quad \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$ è 0 se e solo se i vettori sono complanari

V

$v=(1,1,1)$ e $w=(2,1,-1)$

Trovare un vettore perpendicolare a v e w di modulo 2

$v' = v \wedge w$ $\quad \frac{2}{|v \wedge w|}(v \wedge w)$

$|v'|$

↳ se invece fosse "Trovare $\pm$ vettori":

$v \wedge w = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$

$(0, -(1-2), 1-2)$

$|v \wedge w| = \sqrt{1+1} = \sqrt{2}$

$\pm \frac{2}{\sqrt{2}}(0,1,-1) = \pm\left(0, \frac{2}{\sqrt{2}}, -\frac{2}{\sqrt{2}}\right)$

Calcolare l'angolo tra $v=(2,-2,1)$ e $v'=(0,1,-1)$

$v \cdot v' = 2 \cdot 0 + (-2) \cdot 1 + 1 \cdot (-1) = -3 = |v| \cdot |v'| \cdot \cos\theta$

$|v| = \sqrt{4+4+1} = 3$

$|v'| = \sqrt{1+1} = \sqrt{2}$

$\cos\theta = \frac{-3}{3\sqrt{2}} = -\frac{\sqrt{2}}{2}$

$\theta = \frac{3}{4}\pi$



$v=(1,2,1)$ $\quad v'=(2,0,-1)$

Cerchiamo un vettore non complanare con v e v'

Dati $A=(1,0,2)$ e $B=(3,1,0)$, trovare un punto C sull'asse z tale che il triangolo ABC abbia area 10

↳ $10 = \frac{1}{2} |(C-A) \wedge (B-A)|$

$C=(0,0,z)$ $\quad C-A=(-1,0,z-2)$ $\quad B-A=(2,1,-2)$

$(C-A) \wedge (B-A) = (2-z, -(2-z(z-2)), -1) = (2-z, 2z-6, -1)$

↳ $|(C-A) \wedge (B-A)|^2 = (2-z)^2 + (2z-6)^2 + 1$

cerchiamo z tale che $(2-z)^2 + (2z-6)^2 + 1 = 400$

$A=(1,0,0)$ $\quad B(2,1,1)$ $\quad C(1,-1,0)$ $\quad D(0,1,0)$

Questi punti sono complanari?



B-A = (1, 1, 1)

C-A = (0, -1, 0)

D-A = (-1, 1, 0)

$$\begin{vmatrix} 1 & 1 & 1 \\ 0 & -1 & 0 \\ -1 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 0 & -1 \\ -1 & 1 \end{vmatrix} = -1 \neq 0$$

$$(B-A) \cdot \left((C-A) \wedge (D-A) \right)$$

$5z^2 - 2\delta z - 35y = 0$

$$x_{1,2} = \frac{-\frac{b}{2} \pm \sqrt{\Delta_4}}{a}$$

$\frac{\Delta}{4}$

$\left(\frac{\Delta}{4}\right) = b^2 - 4ac$

$= \frac{b^2}{2} - ac$